

Roll Number:_____

Thapar Institute of Engineering & Technology
Department of Computer Science and Engineering
MID SEMESTER EXAMINATION

B. E. (2nd Yr. COE/CSE)

19th Mar, 2026

Time: 11:30 am to 1:30 pm

Max Marks: 40 (Weightage: 30)

Course Code: UCS310

Course Name: Database Management System

Name of Faculty: Dr. Simranjit (C6), Dr. Shashank (C3), Dr. Amrita (C1), Dr. Mahak (Q1, Q2), Dr. Shivani (C8, C5, Q4), Dr. Ananya (Q3, C7), Dr. Tanya (C2, C4), Dr. Manisha (P1)

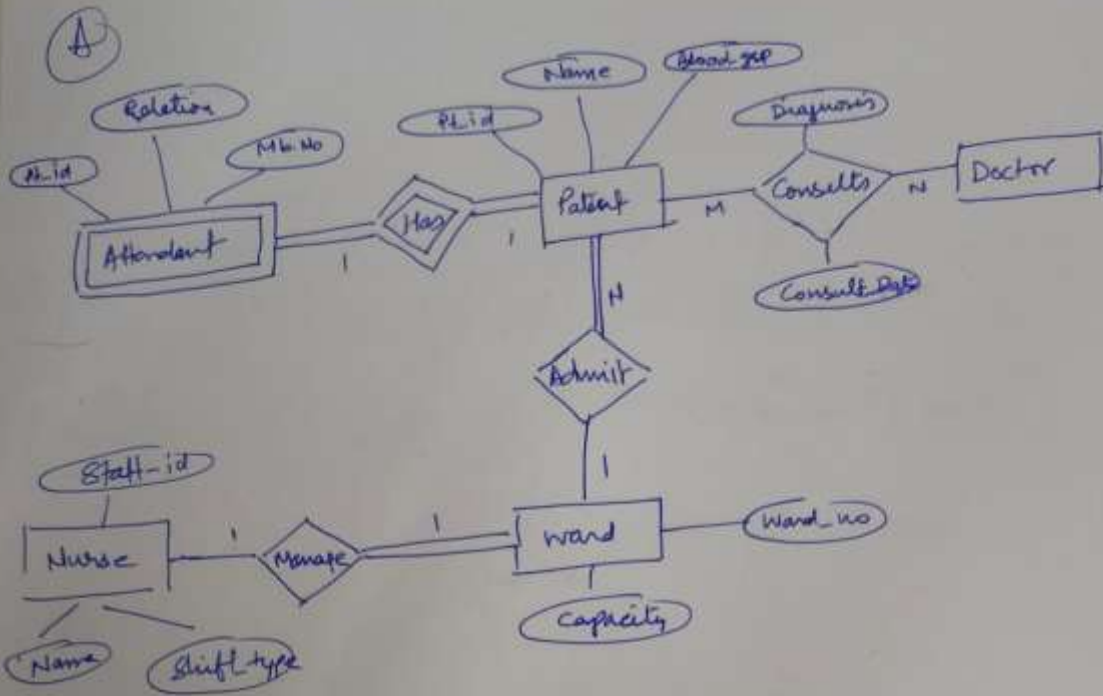
Note: Attempt all Questions in sequence. Answer all sub-parts of each question at one place. Do mention Page No. of your attempt at front page of your answer sheet. Assume missing data (if any).

[illegible]

(c)		
	File-Based System	DBMS
Data Consistency	Low consistency due to multiple files and unsynchronized updates (0.5 mark)	High consistency through centralized control and constraints (0.5 mark)
Data Redundancy	High redundancy due to data duplication across files (0.5 mark)	Reduced redundancy using normalization and integrated database (0.5 mark)
Concurrency Handling	Poor handling; simultaneous access may cause conflicts (0.5 mark)	Efficient handling using transactions, locking, and concurrency control mechanisms (0.5 mark)
Data Integrity	Weak integrity; no strict enforcement of rules (0.5 mark)	Strong integrity using constraints like primary key, foreign key, and validation rules (0.5 mark)

Q.2	The CityHealth Hospital in Bengaluru is transitioning from a legacy database system to a centralized Hospital Information Management System to enhance efficiency and support sustainable development in healthcare through improved data management and resource optimization. You have been hired to gather system requirements and design the database for this new system. As part of the task, you are required to develop a conceptual ER model and map it to a minimized relational schema, while adhering to the following three rules. I. The hospital is divided into several Wards (Ward_Number, Capacity). Each Ward must be managed by exactly one Nurse acting as the In_charge Nurse. Nurses are identified by Staff_ID, Name, and Shift_Type. While every Ward requires a In_charge Nurse, only a small fraction of the nursing staff are In_charge Nurses. A Patient is admitted to a specific Ward. II. The hospital registers Patients (Patient_ID, Name, and Blood_Group). Every Patient must have at least one Attendant recorded. An attendant is not treated as an independent person in the system. They are only identified by their Relationship (e.g. Spouse, Sibling etc.). If a patient's record is deleted, their Attendant data must also be deleted. III. Doctors (Doc_ID, Name, and Specialty) treat multiple Patients. Similarly, a Patient with a complex illness will also be treated by multiple Doctors during their hospital visit. Whenever a doctor sees a Patient, the hospital billing system must record the specific Consult_Date and the resulting Diagnosis for that exact visit. (a) Draw the complete ER Diagram for the CityHealth system with all the cardinality ratios and participation constraints. (b) Map the ER Diagram into the minimum possible number of relational tables. Determine the primary key and foreign key of each table after minimization. Solution: 4 marks (2 marks for diagram + 1 mark for cardinality + 1 mark for participation) for diagram 6 marks (1 mark for each of the six tables with correct primary Key and foreign key)	(10)	CO2	L4
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- ① Attendant (Pt_id, Relationship, Mb.No)
- ② Patient (Pt_id, Name, Blood_grp, Ward-No)
P.K. FK
- ③ consults (Doc_id, Pt_id, Consult_date, Diagnosis)
P.K. (F.K: - Doc_id, Pt_id)
- ④ Doctor (Doc_id, Name, Speciality)
P.K.
- ⑤ Ward (Ward-No, Capacity)
P.K.
- ⑥ Nurse (Staff_id, Name, Shift_id)
P.K.

Q.3	A hospital is developing a digital health information system to monitor patient treatments, doctors, nurses, and ward utilization so that quality healthcare can be delivered efficiently and resources can be managed responsibly. To support this system, the hospital maintains treatment-related information in the following relation: TREATMENT (PatientID, PatientName, DoctorID, DoctorName, Specialization, WardNo, WardType, TreatmentID, TreatmentName, Cost, Date, NurseID, NurseName) The following functional dependencies are known: PatientID → PatientName, DoctorID → DoctorName, Specialization, WardNo → WardType, TreatmentID → TreatmentName, Cost, NurseID → NurseName, (PatientID, TreatmentID, Date) → DoctorID, WardNo, NurseID (a) Determine all candidate keys of the relation with proper justification. (b) Identify and explain the partial and transitive dependencies present in the schema. (c) Normalize the relation up to Third Normal Form (3NF), clearly showing each decomposition step. (d) Examine whether the resulting relations satisfy Boyce-Codd Normal Form (BCNF). If violations exist, perform appropriate decomposition. Solution: (a) Determination of Candidate Key(s) (1 mark) We compute closure of: {PatientID, TreatmentID, Date}⁺ Using FD (6): → DoctorID, WardNo, NurseID Using FD (2): DoctorID → DoctorName, Specialization Using FD (3): WardNo → WardType Using FD (4): TreatmentID → TreatmentName, Cost Using FD (5): NurseID → NurseName Using FD (1): PatientID → PatientName Thus, the closure contains all attributes of the relation. No proper subset of this combination can determine all attributes. Candidate Key = (PatientID, TreatmentID, Date) (b) Identification of Partial and Transitive Dependencies (3 mark)	(10)	CO3	L3
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Partial Dependencies [1 Marks {0.5 Marks for each partial dependencies}] A partial dependency occurs when a non-prime attribute depends on part of a composite key. Since the key is (PatientID, TreatmentID, Date): ● PatientID → PatientName (depends on part of key) ● TreatmentID → TreatmentName, Cost (depends on part of key) These violate 2NF. Transitive Dependencies[2 Marks {0.5 Marks for each transitive dependency}] A transitive dependency occurs when: Key → Non-key → Another Non-key Examples: ● (PatientID, TreatmentID, Date) → DoctorID ● DoctorID → DoctorName, Specialization ● (PatientID, TreatmentID, Date) → WardNo WardNo → WardType ● (PatientID, TreatmentID, Date) → NurseID NurseID → NurseName These violate 3NF. (c) Normalization up to Third Normal Form (3NF) (3 Marks) Step 1: Convert to 2NF (Remove Partial Dependencies) Separate attributes depending only on PatientID and TreatmentID. PATIENT Table PATIENT(PatientID, PatientName) TREATMENT_MASTER Table TREATMENT_MASTER(TreatmentID, TreatmentName, Cost) TREATMENT_RECORD Table TREATMENT_RECORD(PatientID, TreatmentID, Date, DoctorID, WardNo, NurseID) Step 2: Remove Transitive Dependencies (Convert to 3NF)			
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Now decompose based on remaining FDs: DoctorID → DoctorName, Specialization WardNo → WardType NurseID → NurseName DOCTOR Table DOCTOR(DoctorID, DoctorName, Specialization) WARD Table WARD(WardNo, WardType) NURSE Table NURSE(NurseID, NurseName) Final 3NF Schema 1. PATIENT(PatientID, PatientName) 2. TREATMENT_MASTER(TreatmentID, TreatmentName, Cost) 3. DOCTOR(DoctorID, DoctorName, Specialization) 4. WARD(WardNo, WardType) 5. NURSE(NurseID, NurseName) 6. TREATMENT_RECORD(PatientID, TreatmentID, Date, DoctorID, WardNo, NurseID) All non-key attributes now depend only on the key. (d) Verification of BCNF (3 marks) BCNF requires: For every FD $X \rightarrow Y$, X must be a super key. Check each relation:			
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	Relation	FD	Determinant is Key?	BCNF			
	PATIENT	PatientID → PatientName	Yes	✓			
	TREATMENT_MASTER	TreatmentID → TreatmentName, Cost	Yes	✓			
	DOCTOR	DoctorID → DoctorName, Specialization	Yes	✓			
	WARD	WardNo → WardType	Yes	✓			
	NURSE	NurseID → NurseName	Yes	✓			
	TREATMENT_RECORD	(PatientID, TreatmentID, Date) determines rest	Yes	✓			
	All relations satisfy BCNF. No further decomposition required.						
Q.4	(a) An eco-friendly retail company is developing a data management system to monitor product sales through both online platforms and physical stores. The goal is to analyze product demand, reduce overproduction, and manage inventory efficiently to minimize waste and promote sustainable retail practices. The company stores product and order information in the following relational schema: Products (ProductID, ProductName, CategoryID, Price), OnlineOrders (OrderID, ProductID) and StoreOrders (OrderID, ProductID). Write SQL queries to fetch following details from database: i) Display CategoryID and the number of products in each category that contains more than four products. ii) List all products and their online order IDs, including products with no online orders. iii) Show product IDs and order IDs from OnlineOrders and StoreOrders such that the result includes products appearing only in online orders as well as those appearing only in store orders. iv) Display ProductID, ProductName, Online OrderID, and Store OrderID for products that have been ordered both online and in-store. v) Display products that have been ordered either online or in-store, and whose price is greater than the average price of products in the same category. Solution: i) SELECT CategoryID, COUNT(ProductID) AS TotalProducts (1 mark) FROM Products GROUP BY CategoryID HAVING COUNT(ProductID) > 4; ii) SELECT P.ProductID, P.ProductName, O.OrderID (1 mark) FROM Products P LEFT JOIN OnlineOrders O ON P.ProductID = O.ProductID;				(7)	C05	L3

	iii) SELECT O.ProductID, O.OrderID AS OnlineOrderID, S.OrderID AS StoreOrderID (1 mark) FROM OnlineOrders O FULL OUTER JOIN StoreOrders S ON O.ProductID = S.ProductID; iv) SELECT P.ProductID, P.ProductName, O.OrderID AS OnlineOrderID, S.OrderID AS StoreOrderID FROM Products P INNER JOIN OnlineOrders O ON P.ProductID = O.ProductID INNER JOIN StoreOrders S ON P.ProductID = S.ProductID; (1 mark) v) SELECT ProductID, ProductName, Price (3 marks) FROM Products p WHERE Price > (SELECT AVG(Price) FROM Products WHERE CategoryID = p.CategoryID) AND EXISTS (SELECT * FROM (SELECT ProductID FROM OnlineOrders UNION SELECT ProductID FROM StoreOrders) WHERE o.ProductID = p.ProductID); (b) Suppose we have a database for an investment firm, consisting of the following attributes: B – Broker, O – Office of a broker, I – Investor, S – Stock, Q – Quantity of stock owned by an investor, D – dividend paid by a stock. Hence, the overall schema is R = (B, O, I, S, Q, D). Assume that the following functional dependencies are required to hold this database. $I \rightarrow B, \quad IS \rightarrow Q, \quad B \rightarrow O, \quad S \rightarrow D$ On the basis of given information, write true or false for the following statements: i) If R is decomposed into R1(I, B), R2(B, O), and R3(I, S, Q, D), the decomposition is lossless. ii) If R is decomposed into R1(I, B, O) and R2(I, S, Q, D), the decomposition is lossless but not dependency preserving. iii) If a decomposition preserves all functional dependencies, it automatically guarantees a lossless join. Solution: i) TRUE (1 mark) ii) False (1 mark) iii) False (1 mark)			L2 (3) CO3
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